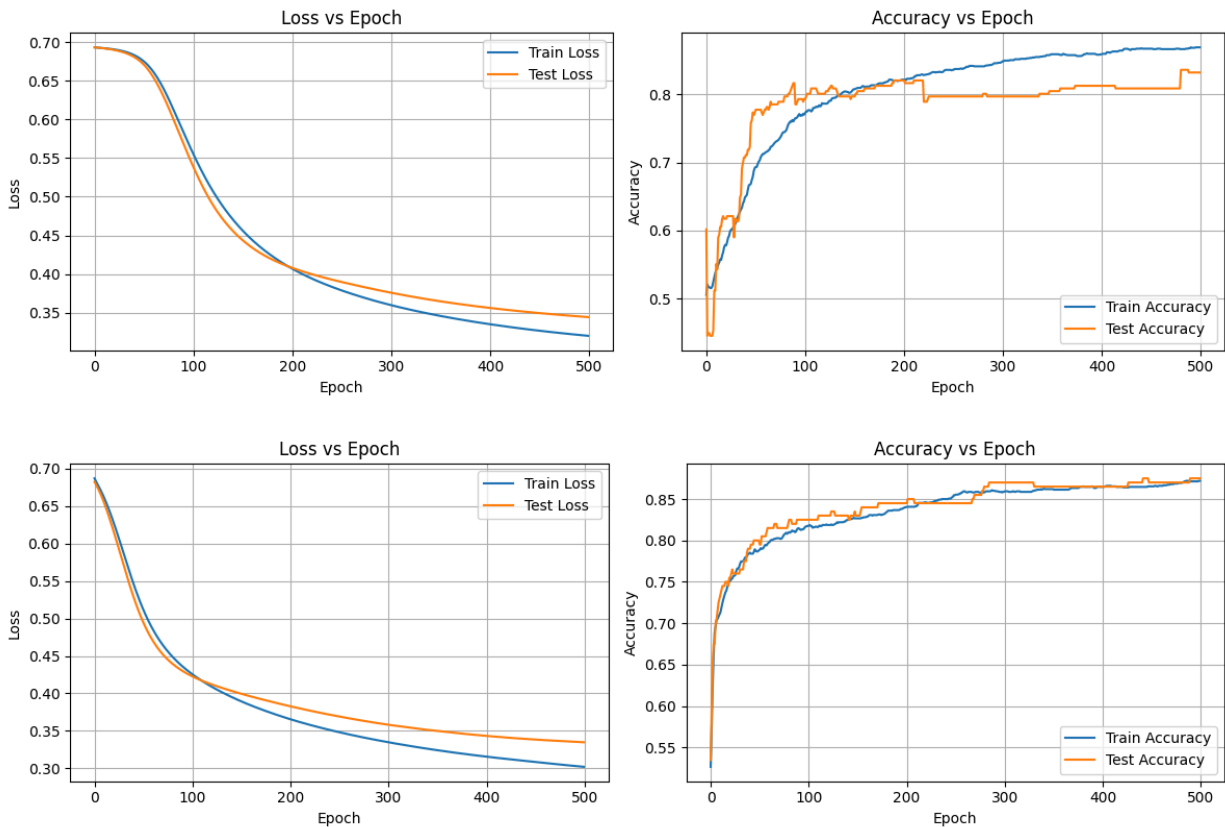


Report on Loss and Accuracy between Manual implementation and Pytorch implementation of a single layer NN



(top row): results from [nn.py](#)

(bottom row): results from [reference.py](#)

Analysis:

Examining the results from the manual implementation, The loss vs epoch function looks more like a logistic model, At close to Epoch 200, the training and testing loss converge, but beyond that, the rate of which the test data's loss decreases, decreases in comparison to the training data's rate. When looking at the accuracy of the manual interpretation, initially, there is a major drop in the test data's accuracy, but that gets quickly reminded, and the line more or less follows or remains parallel to the training data's accuracy.

Examining the result of pytorch's implementation, the plot of the training and testing data's losses follow more of an exponential curve and there is a convergence that occurs at approximately Epoch 100 and the test data's curve/ rate of descent beyond the convergence does slow down in comparison to the training data. Looking at the accuracies of the data sets however, the test data more closely resembles the training data.

Between the 2 implementations of the single layer network, though both do perform relatively similar, Pytorch is considerably steadier compared to my implementation when it comes to accuracy and has a better loss.